Three-generator hypertopes of prescribed two-power order

Human Compression Project

Abstract

We recover the complete three-generator construction of Hou, Feng and Leemans: its defining presentation, exact order and type, regular hypertope, local cyclic kernels and quotient structure. Three dihedral attachments to an affine square quotient give the group. The normal-kernel recognition theorem identifies the source presentation without a finite-order comparison. The parity correction in the printed structural display remains explicit.

1 Marked groups and geometric scope

A marking is a generating list of nonidentity involutions. For a subset J of its labels, write G_J for the subgroup generated by those markers. The intersection property requires $G_J \cap G_K = G_{J\cap K}$. In rank three the geometric type used here records all three pair-product orders. A regular hypertope is a thin, residually connected incidence geometry whose type-preserving automorphism group is transitive on chambers. The marking need not be string. Thus its group intersection property alone is insufficient for the geometric assertion.

We use the complete companion proofs UA.SK and UA.SK.Geometry. The former constructs a group by separately supported local kernels. The latter also transfers the typed coset intersection property from a regular-hypertope base and proves that the constructed involutory marking gives the full regular-hypertope group. All its hypotheses are checked below. Write D_m for the dihedral group of order m, and use $[x, y] = x^{-1}y^{-1}xy$. The exact source edition is arXiv:1901.09034v1, dated 23 January 2019; its principal construction is Theorem 3.2. The auxiliary square decompositions are also supplied by UA.SquareDecomposition, with the earlier attribution acknowledged in the source's Lemma 3.1.

2 Affine seeds and the source presentation

We retain the proved affine square model [UA, UA.Affine, UA.Square and UA.SquareDecomposition]. In coordinates, with the rightmost transformation acting first, it is

$$a(x, y) = (-x, y), \qquad b(x, y) = (y, x), \qquad c(x, y) = (x, 1 - y).$$

These generate $W = \mathbb{Z}^2 \rtimes D_8$ with complete presentation

$$\langle a, b, c \mid a^2 = b^2 = c^2 = (ab)^4 = (bc)^4 = (ac)^2 = 1\rangle. \tag{1}$$

For $\delta \geq 0$, define

$$Q_\delta = \begin{cases} W/q\mathbb{Z}^2, & \delta \text{ even}, \quad q = 2^{(\delta+2)/2}, \\ W/q\langle(1, 1), (1, -1)\rangle, & \delta \text{ odd}, \quad q = 2^{(\delta+1)/2}. \end{cases}$$

It has order $2^{\delta+5}$, pair orders $(4, 4, 2)$ and all Boolean deletions. To identify the parity relation exactly, use $[x, y] = x^{-1}y^{-1}xy$. Direct affine multiplication gives translations

$$[(ab)^2, c] = (0, -2), \qquad [(ab)^2, (bc)^2] = (-2, -2).$$

The normal closure of a translation in W is the lattice spanned by its signed-permutation orbit. Hence, respectively, adding

$$[(ab)^2, c]^{2^{\delta/2}} = 1 \quad \text{or} \quad [(ab)^2, (bc)^2]^{2^{(\delta-1)/2}} = 1 \tag{2}$$

to (1) gives exactly Q_δ.

Take $n \geq 10$, $s, t \geq 2$, $\ell \geq 1$, and $\delta = n - s - t - \ell \geq 0$. Let P have the three involutory markers, the pair-power relations

$$(ab)^{2^s} = (bc)^{2^t} = (ac)^{2^\ell} = 1,$$

the three relations

$$[(ab)^4, c] = [a, (bc)^4] = [(ac)^2, b] = 1, \tag{3}$$

and the appropriate parity relation (2). This is the defining presentation of [HFL19, Theorem 3.2], with renamed markers.

Theorem 1 (Hou–Feng–Leemans, corrected structural display)**.** *The presentation P is the supported extension of Q_δ by the three marked dihedral quotients of orders $2^{s+1}, 2^{t+1}, 2^{\ell+1}$ onto its pair subgroups. In particular it has order 2^n, the prescribed pair orders, and a regular rank-three hypertope with full type-preserving group P. The subgroups*

$$K_{ab} = \langle (ab)^4 \rangle, \qquad K_{bc} = \langle (bc)^4 \rangle, \qquad K_{ac} = \langle (ac)^2 \rangle$$

form its direct-product base kernel, of type $C_{2^{s-2}} \times C_{2^{t-2}} \times C_{2^{\ell-1}}$.

Proof. The three pair supports are an antichain. Their proper marked subgroups are the trivial group and the reflection subgroups, so the maps to the base are injective on every proper marked subgroup. UA.SK therefore constructs H.

All relators of P hold in H. Each pair-power kernel word is supported only in its own coordinate, where the third marker is the identity, proving (3). The two commutator words in (2) vanish in each local pair coordinate: one of the relevant markers is deleted, and direct substitution gives the identity. Their base values are the specified translations. Each local dihedral presentation holds in P. Within its pair, each marker inverts the displayed kernel generator; the third marker centralizes it by (3). Thus all three K's are normal. Killing them reduces the presentation precisely to (1) and (2). UA.Recognition applies.

Its order formula gives $2^{\delta+5}2^{s-2}2^{t-2}2^{\ell-1} = 2^n$ and its local recognition gives the exact pair orders. The square base is a regular string polytope; its maximal-parabolic cosets have the needed flag property by the retained polytope correspondence. The dihedral local markers are involutions, so UA.SK.Geometry supplies the claimed hypertope and its full group. This last use is explicit: intersection property alone does not supply the flag property of a general coset geometry. □

The quotient description is consequently

$$P \cong (C_{2^{s-2}} \times C_{2^{t-2}} \times C_{2^{\ell-1}}) \,.\, \begin{cases} (D_{2^{(\delta+4)/2}} \times D_{2^{(\delta+4)/2}}) \rtimes C_2, & \delta \text{ even}, \\ (D_{2^{(\delta+3)/2}} \times D_{2^{(\delta+3)/2}}) \rtimes C_2^2, & \delta \text{ odd}. \end{cases}$$

Here the extension dot makes no splitting assertion. These are the corrected parity rows already recorded in the accepted source contract; the source interchanges them and repeats the first pair word when naming its second kernel. The present argument preserves both corrections. The decompositions of the square quotient are the previously proved UA.SquareDecomposition, not new group classifications.

3 Attribution and exact boundary

The construction and its geometric conclusion are the source result. The proof uses one normal supported-kernel recognition theorem for all parameters. Its local obligations are the three dihedral maps, the literal parity relator, the normality of the three kernel subgroups, and exact identification of the residual square quotient. Those checks have all been made; no cardinality estimate for the unrecognized presented group is used. The affine normal form, both lattice quotients and both structural decompositions have complete proofs in the companion umbrella. Its regular-polytope correspondence and typed-Helly geometry interface are explicit dependencies.

This is the full admitted principal construction, with the source's two structural corrections. It does not absorb the separate conjectural nonexistence statement numbered 3.3 into a construction theorem. Any answer to that question has its own proof and provenance. The present paper makes no claim about correction priority or another source edition.

References

[1] [UA] *Supported kernels and finite two-group quotients*, companion umbrella in this volume.

[2] [HFL19] D.-D. Hou, Y.-Q. Feng and D. Leemans, *Existence of regular 3-hypertopes with* 2^n *chambers*, arXiv:1901.09034v1, 23 January 2019; published in *Discrete Mathematics* 342 (2019), 1857–1863.

[3] [MS02] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge University Press, 2002.